Assignment - 2

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course code - MLA0205

course Name - fundamentals of machine learning.

Task-1 :- Analysis of 5 Machine learning systems.

a) loan Default prediction.

* learning paradigm:- supervised learning.

* learning task :- classification (predicting whether a customer will default or not).

* Experience (E):- Ten years of labelled loan repayment records.

* performance measure (P):- Accuracy, precision, recall and f1- score.

b) customer Behaviour segmentation:-

Learning paradigm :- unsupervised learning.

* learning task (T):- clustering (grouping) customers).

* Experience (E):- Data of 2 million customers without performance (P):- silhouette score and cluster quality.

c) credit - card spending prediction:-

learning pradigm :- supervised learning.

* learning task (T):- predict next month's spending

* Experience (E):- previous credit - card spending records.

* performance (P):- previous credit - card spending records.

d). ATM cash- refill policy:-

T:- Decide when and how much cash to refill

P:- Maximum cash availability with minimum transport cost

E:- Repeated trials and feedback from ATM operations.

e) fraud detection:-

T:- Identify fraudulent and genuine transcar

P:- Accuracy, precision and recall.

E:- 2% labelled and 98% unlabelled transcar

Why system is most affected by label noise

loan default prediction is highly affected by label noise because incorrect repayment labels can teach the model wrong patterns.

Task-2:- probability of disease detection.

Given

disease prevalence = 2% = 0.02

sensitivity = 95% = 0.95

specificity = 90% = 0.90

false positive rate = 10% = 0.10

using Bayes theorem:-

$$P(D|+) = \frac{P(+|D)\,P(D)}{P(+)}$$

$$= \frac{0.95 \times 0.02}{(0.95 \times 0.02) + (0.10 \times 0.98)}$$

$$= \frac{0.019}{0.117} = 0.1624 = 16.24\%.$$

The probability that a person actually has the disease after testing positive is 16.24%.

Analysis:-

The answer is much lower than sensitivity because only 2% of the population has the disease

Task-3 :- joint probability distribution

Given

X/Y	Y=0	Y=1
X=0	0.10	0.20
X=1	0.15	0.25
X=2	0.20	0.10

a) find marginal probabilities :-

$p(x=0) = 0.10+0.20 = 0.30$

$p(x=1) = 0.15+0.25 = 0.40$

$p(x=2) = 0.20+0.10 = 0.30$

$p(y=0) = 0.10+0.15+0.20 = 0.45$

$p(y=1) = 0.20+0.25+0.10 = 0.55.$

b) calculate $E(x)$, $Var(x)$ and $E(y)$.

$E(x) = \sum x\,p(x)$

$= 0(0.30)+1(0.40)+2(0.30)$

$= 1.$

$E(x^2) = 0^2(0.30)+1^2(0.40)+2^2(0.30)$

$= 0+0.40+1.20 = 1.60$

$Var(x) = E(x^2)-(E(x))^2$

$= 1.60-1^2 = 0.60$

$E(y) = 0(0.45)+1(0.55) = 0.55.$

c) calculate conditional probability.

$$p(x=2/y=1) = \frac{p(x=2, y=1)}{p(y=1)}$$

$$= \frac{0.10}{0.55} = 0.1818$$

d) calculate covariance & correlation

$$E(XY) = 0+0+0.25+0.80 = 0.45$$

$$COV(X,Y) = E(XY) - E(X)E(Y)$$

$$= 0.45 - 1 \times 0.55$$

$$= -0.10$$

$$r = \frac{COV(X,Y)}{\sqrt{Var(X)Var(Y)}}$$

$$Var(Y) = 0.55(0.45) = 0.2475$$

$$r = \frac{-0.10}{\sqrt{0.60 \times 0.2475}}$$

$$r = -0.2595.$$

e) Are X and Y independent

$$P(X,Y) = P(X)P(Y)$$

$$P(0,0) = 0.10$$

But $P(X=0)\,P(Y=0) = 0.30 \times 0.45 = 0.135$

since 0.10 is not equal to 0.135.

Task-4 :- Normal distribution and Maximum likelihood estimation.

a) calculate probabilities.

$$X \sim N(50, 16)$$

$$\mu = 50, \quad \sigma = \sqrt{16} = 4, \quad z = \frac{x-\mu}{\sigma}$$

i. find $P(X > 58)$

$$z = \frac{58-50}{4} = 2$$

$$P(X>58) = 1 - \Phi(2) = 1 - 0.9772 = 0.0228$$

2. find $p(46 < x < 54)$

$$z_1 = \frac{46-50}{4} = -1, \quad z_2 = \frac{54-50}{4} = 1.$$

$$p(-1 < z < 1) = 0.8413 - 0.1587 = 0.6826$$

b) Maximum likelihood estimates.

Given

12, 15, 11, 14, 13.

step:-1 calculate mean

$$\mu = \frac{12+15+11+14+13}{5} = 13.$$

step-2:- calculate variance

x	x-Mean	$(x-\text{Mean})^2$
12	-1	1
15	2	4
11	-2	4
14	1	1
13	0	0

$$\sigma^2_{ML} = \frac{\sum (x-\mu)^2}{n} \qquad s^2 = \frac{\sum (x-\mu)^2}{n-1}$$

$$= \frac{10}{5} \qquad = \frac{10}{4}$$

$$= 2 \qquad = 2.5$$

ML variance is 2, while unbiased variance is 2.5 ML

Task-5:- Entropy and average code length.

Given

$$p = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right)$$

a) calculate entropy

$$H(x) = -\sum p(x) \log_2 p(x)$$

$$= -(0.5 \log_2 0.5 + 0.25 \log_2 0.25 + 0.125 \log_2 0.125 + 0.125 \log_2 0.125)$$

$$= 0.5 + 0.5 + 0.375 + 0.375$$

$$H(x) = 1.75$$

b) compare with uniform source

$$H_{max} = \log_2 4$$

$$= 2 \text{ bits}$$

$$R = H_{max} - H(x)$$

$$= 2 - 1.75$$

$$= 0.25 \text{ bit}$$

percentage redundancy $= \frac{0.25}{2} \times 100 = 12.5\%$

c) Average code length

Given

0, 10, 110, 111

code lengths are 1, 2, 3, 3

$$L = \sum p(x) \cdot l(x)$$

$$= 0.5(1) + 0.25(2) + 0.125(3) + 0.125(3)$$

$$L = 1.75 \text{ bits.}$$

The average code length equals the entropy.

Task-6 :- KL - Divergence

$$P = (0.5, 0.25, 0.25)$$

$$Q = (1/3, 1/3, 1/3)$$

a) calculate KL (P||Q).

$$D_{KL}(P||Q) = \sum p(x) \log_2 \frac{p(x)}{Q(x)}$$

$$= 0.5 \log_2(1.5) + 0.25 \log_2 0.75 + 0.25 \log_2 0.75$$

$$= 0.085 \text{ bits}$$

b) calculate $KL(Q||P)$.

$$D_{KL}(Q||P) = \sum Q(x) \log_2 \frac{Q(x)}{P(x)}$$

$$= \frac{1}{3} \log_2 \frac{1/3}{0.5} + \frac{1}{3} \log_2 \frac{1/3}{0.25} + \frac{1}{3} \log_2 \frac{1/3}{0.25}$$

$$= 0.0817 \text{ bits}$$

c) Analysis

KL divergence measures the difference b/w two probability distribution

*KL divergence is not symmetric

* $KL(P||Q)$ is not equal to $KL(Q||P)$.

Task-7 :- Entropy and mutual information

A	B	probability
0	0	0.4
0	1	0.1
1	0	0.1
1	1	0.4

a) calculate $H(A)$ & $H(B)$.

$$P(A=0) = 0.5, \quad P(A=1) = 0.5$$

$$P(B=0) = 0.5, \quad P(B=1) = 0.5$$

$$H(A) = -(0.5 \log_2 0.5 + 0.5 \log_2 0.5) = 1 \text{ bit}$$

$$H(B) = 1 \text{ bit}$$

b. calculate joint entropy.

$$H(A,B) = -\sum p(A,B) \log_2 p(A,B)$$

$$= -(2(0.4 \log_2 0.4) + 2(0.1 \log_2 0.1))$$

$$H(A,B) = 1.722 \text{ bits}$$

c) calculate conditional entropy.

$$H(A|B) = H(A,B) - H(B)$$

$$= 1.722 - 1$$

$$\approx 0.722$$

d) calculate mutual information

$$I(A,B) = H(A) - H(A|B)$$

$$= 1 - 0.722$$

$$\approx 0.278 \text{ bits}$$

e) Analysis:

Mutual information measures how much information

B gives about A.

* value of 0.278 bits

TASK - 8 :- Bayes Decision theory.

$p(x/\omega_1) = N(2,4)$ $\quad p(\omega_1) = 0.6$

$p(x/\omega_2) = N(6,4)$ $\quad p(\omega_2) = 0.4$

a) Derive Bayes decision threshold

$$p(\omega_1)p(x|\omega_1) = p(\omega_2)p(x|\omega_2)$$

$$0.6 \exp\left(-\frac{(x-2)^2}{8}\right) = 0.4 \exp\left(-\frac{(x-6)^2}{8}\right)$$

$$\ln\left(\frac{0.6}{0.4}\right) = \frac{(x-2)^2 - (x-6)^2}{8}$$

$$x^* = 4 + \ln(1.5) \approx 4.4055$$

b) classify $x = 4.5$

$4.5 > 4.4055$

ω_2

c) overall probability of error

$P(\text{error}) = 0.6\, p(x > x^* | \omega_1) + 0.4\, p(x < x^* | \omega_2)$

$P(\text{error}) \simeq 0.1538$

$P(\text{error}) = 15.38\%$

d) Effect of increasing prior probability.

When the prior probability of class ω_1 increases, the decision threshold moves towards right

Task-9:- Minimum Risk Decision.

$P(\omega_1 / x) = 0.3$

$P(\omega_2 / x) = 0.7$

a) calculate conditional risk

$R(\alpha_1 / x) = 0(0.3) + 8(0.7)$

$= 5.6$

$R(\alpha_2 / x) = 50(0.3) + 0(0.7)$

$= 15.$

b) Minimum-risk action

$5.6 < 15$

α_1 = Declare disease

c) Find the threshold

$P(\omega_1 / x) = P$

$R(\alpha_1) = 8(1-P)$

$R(\alpha_2) = 50P$

$$8(1-p) = 50p$$

$$8 = 58p$$

$$p = \frac{8}{58}$$

$$p \approx 0.1379 = 13.79\%$$

Task-10 Bass-variance Trade-off

a) calculate generalisation gap

Gap: validation MSE - Training MSE

$= 9.5 - 9.0$

$= 0.5$

b) Diagnose underfitting and overfitting

Degree-1:- underfitting because the model is too simple

Degree-2:- Better performance than degree-1.

c) select the best model.

Degree-3

It has the lowest validation MSE of 2.9

It balances bias & variance

d) Two techniques to improve degree-9

1. Regularization - use L_1 or L_2 regularization to reduce model

2. Cross-validation : use cross-validation to select to a suitable polynomial degree

TASK-11:- Bayesian estimation

No. of tosses = 10, Heads = 7, Tails = 3

a) calculate ML estimate

$$\theta_{ML} = \frac{\text{Heads}}{\text{Total tosses}}$$

$$= \frac{7}{10} = 0.7$$

b) calculate MAP estimate

$$\theta_{MAP} = \frac{a+H-1}{a+b+N-2}$$

$$= \frac{2+7-1}{2+2+10-2} = \frac{8}{12}$$

$$= 0.6667$$

c) calculate posterior mean

$$E(\theta) = \frac{a+H}{a+b+N}$$

$$= \frac{2+7}{2+2+10} = \frac{9}{14}$$

$$= 0.6429$$

d) If 40 head occur in 100 tossed

Estimate	Formula	Answer
ML	40/100	0.7000
MAP	71/104	0.6827
posterior mean	72/104	0.6923

Task-12:- Design a sepsis early-learning system

a) Design the learning system

1. Task (T):- predict whether an ICU patient will develop sepsis within 6 hours.

2. performance measure (P):-

Accuracy, sensitivity, specificity

b) calculate TP, FN, FP, TN.

$TP = 50 \times 0.90 = 45$

$FN = 50 - 45 = 5$

$TN = = 950 \times 0.80 = 760$

$FP = 950 - 760 = 190$

c) calculate PPV & NPV.

$$PPV = \frac{TP}{TP+FP} = \frac{45}{45+190}$$

$= 19.15\%$

$$NPV = \frac{TN}{TN+FN}$$

$$= \frac{760}{760+5}$$

$$= \frac{760}{765}$$

$= 99.35\%$.